LAB 7: Fuzzing for Input Validation Bugs

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Run the buggy script

```
PS C:\Users\ANONYMOUS\OneDrive\Docs\L6_L7_47_fuel_station\Lab 7> python fuel_station.py
==== Running parse_purchase ====
Enter grade:litres (e.g. diesel:25): 25
Traceback (most recent call last):
  File "C:\Users\ANONYMOUS\OneDrive\Docs\L6_L7_47_fuel_station\Lab 7\fuel_station.py", line 31, in <module>
    main()
    ~~~~^^
  File "C:\Users\ANONYMOUS\OneDrive\Docs\L6_L7_47_fuel_station\Lab 7\fuel_station.py", line 17, in main
    result = parse_purchase(user_input)
  File "C:\Users\ANONYMOUS\OneDrive\Docs\L6_L7_47_fuel_station\Lab 7\fuel_station.py", line 2, in parse_purchase
    grade, litres = text.split(':')
    ^^^^^^^^^^^^^
ValueError: not enough values to unpack (expected 2, got 1)
PS C:\Users\ANONYMOUS\OneDrive\Docs\L6_L7_47_fuel_station\Lab 7> python fuel_station.py
==== Running parse_purchase ====
Enter grade:litres (e.g. diesel:25): diesel:20
Grade and Litres: ('Diesel', 20.0)

==== Running volume_average ====
Enter pump volumes in litres as CSV (e.g. 10,25,40): 10
Average Volume: 10

==== Running payable_bill ====
Enter bill|discount (e.g. 2500|100): 2000|100
Payable Bill: 2260.0
```

Run the tests against the buggy code

```
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=========================================== short test summary info ============================================
FAILED test_fuel_station.py::test_parse_purchase_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-excep
tions)
FAILED test_fuel_station.py::test_parse_purchase_output_is_valid - ExceptionGroup: Hypothesis found 3 distinct failures. (3 su
b-exceptions)
FAILED test_fuel_station.py::test_volume_average_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-excep
tions)
FAILED test_fuel_station.py::test_volume_average_output_is_valid - ExceptionGroup: Hypothesis found 2 distinct failures. (2 su
b-exceptions)
FAILED test_fuel_station.py::test_payable_bill_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-excepti
ons)
FAILED test_fuel_station.py::test_payable_bill_output_is_valid - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-
exceptions)
========================================== 6 failed, 3 passed in 16.88s ==========================================
PS C:\Users\ANONYMOUS\OneDrive\Docs\L6_L7_47_fuel_station\Lab 7>
```

Fix the script

```
PS C:\Users\ANONYMOUS\OneDrive\Docs\L6_L7_47_fuel_station\Lab 7> python fuel_station.py
==== Running parse_purchase ====
Enter grade:litres (e.g. diesel:25): 20
Grade and Litres: Invalid input - please follow the format shown.

==== Running volume_average ====
Enter pump volumes in litres as CSV (e.g. 10,25,40): l10
Average Volume: Invalid input - please follow the format shown.

==== Running payable_bill ====
Enter bill|discount (e.g. 2500|100): 0
Payable Bill: Invalid input - please follow the format shown.
PS C:\Users\ANONYMOUS\OneDrive\Docs\L6_L7_47_fuel_station\Lab 7> python fuel_station.py
==== Running parse_purchase ====
Enter grade:litres (e.g. diesel:25): diesel:30
Grade and Litres: ('Diesel', 30.0)

==== Running volume_average ====
Enter pump volumes in litres as CSV (e.g. 10,25,40): 300
Average Volume: 300.0

==== Running payable_bill ====
Enter bill|discount (e.g. 2500|100): 2200|100
Payable Bill: 2496.0
PS C:\Users\ANONYMOUS\OneDrive\Docs\L6_L7_47_fuel_station\Lab 7>
```

Re-run the tests

```
PS C:\Users\ANONYMOUS\OneDrive\Docs\L6_L7_47_fuel_station\Lab 7> python -m pytest test_fuel_station.py
============================= test session starts =============================
platform win32 -- Python 3.14.3, pytest-9.2.0, pluggy-1.7.0
rootdir: C:\Users\ANONYMOUS\OneDrive\Docs\L6_L7_47_fuel_station\Lab 7
plugins: anyio-4.15.1, hypothesis-6.170.15
collected 9 items

test_fuel_station.py .........                                           [100%]

============================== 9 passed in 6.59s ==============================
PS C:\Users\ANONYMOUS\OneDrive\Docs\L6_L7_47_fuel_station\Lab 7>
```